KARTIKEYA ARVIND YADAV

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 3+ years of experience in firmware development, RTOS, and hardware-software integration for ARM Cortex and x86 platforms. Expert in C/C++ programming, embedded Linux (Yocto), networking protocols (TCP/IP, UDP, SIP), and BIOS/UEFI customization. Experienced with IoT architectures, AWS cloud integration, and full-stack development using Qt/QML and Angular.

TECHNICAL SKILLS

Programming: C, C++, Python, Embedded C, Assembly, JavaScript, TypeScript
Embedded Systems: FreeRTOS, Zephyr RTOS, Embedded Linux, Yocto BSP, Linux Kernel Modules, Device Drivers, BIOS/UEFI (EDK2)
Hardware: ARM Cortex-M/A, STM32, ESP32, Nordic nRF, UART, SPI, I²C, PCIe
Networking: TCP/IP, UDP, MQTT, SIP, VoIP, Ethernet, BLE, 802.11 Wi-Fi
Web & UI: Angular, Ionic, Qt/QML, React, HTML/CSS, RESTful APIs
Cloud & DevOps: AWS (IoT Core, Lambda, DynamoDB), Docker, CI/CD (Jenkins, GitHub Actions), Git
Tools: JTAG, GDB, Oscilloscope, Logic Analyzer, Wireshark, CMake, QEMU

EDUCATION

Master of Science in Computer Science – Drexel University, Philadelphia, PA – GPA: 3.89 *June 2025*
Bachelor of Technology in Electronics and Communication Engineering – Shiv Nadar University, India *May 2021*

PROFESSIONAL EXPERIENCE

BlackSquirrel Collaborative Inc Philadelphia, PA
Software Engineer April 2026 – Present
- Develop responsive UI components using Angular and Ionic for scalable web application focused on business assessment solutions
- Implement form validation, multilingual capabilities, and cross-device compatibility enhancements
- Collaborate with cross-functional teams following Agile workflows to deliver production-ready features

Handshake AI Solution LLC Remote
Spectra Fellow – AI & Systems Software Development October 2025 – Present
- Design evaluation frameworks for LLM-generated technical content in embedded systems domains

Jitsik LLC Philadelphia, PA
Graduate Co-op Intern September 2024 – December 2024
- Developed and refactored C++/C# embedded firmware for IMU sensor calibration in VR simulators

Smith+Nephew Andover, MA
R&D Software Engineering Intern June 2024 – August 2024
- Built multi-threaded C/Python services for IoT medical devices integrating AWS IoT Core with MQTT protocols under HIPAA compliance

Champion Semiconductor LLP GEOCON India
Embedded Software Engineer January 2021 – May 2023
- Developed production-grade C/C++ firmware for ARM Cortex and x86 platforms with real-time behavior
- Customized Yocto-based Embedded Linux builds integrating BSPs, device drivers, and middleware
- Engineered UEFI/BIOS firmware using EDK2 framework with secure boot implementation

KEY TECHNICAL PROJECTS

IoT-Enabled Surgical Device: Prototyped cloud-connected modules on Zephyr RTOS using Nordic nRF with AWS IoT telemetry under HIPAA compliance
Passenger VoIP System: Developed Qt-based VoIP communication system integrating SIP protocols with REST API for remote management
Driver Fatigue Monitoring: Designed TensorFlow/OpenCV pipeline for real-time facial detection on ARM embedded platforms
Network Video Recorder: Built Qt-based GUI integrating GStreamer pipelines for IP video/audio streaming with network protocol optimization

CERTIFICATIONS & ACHIEVEMENTS

- Inducted into Upsilon Pi Epsilon Honor Society for academic excellence in computing
- Presented driver fatigue monitoring system to Indian Air Force; exhibited solutions at Electronica India 2022